

Pointwise Complexity for Gaussian Fields and Finite-Scale Renormalization

Yunbei Xu
National University of Singapore
yunbei@nus.edu.sg

Abstract

We prove a pointwise majorizing-measure theorem for centered Gaussian processes. Classical generic chaining characterizes the scalar quantity $\mathbb{E} \sup_{x \in T} X_x$; the theorem here gives a simultaneous high-probability envelope for the entire field. For an ambient prior μ , the envelope at x is governed by a localized Fernique–Talagrand functional $\Phi_\mu(x)$, truncated at the variance scale $\sigma(x) = (\mathbb{E} X_x^2)^{1/2}$, together with the corresponding Gaussian tail term. Optimizing the resulting in-expectation bound recovers the anchored Fernique–Talagrand scale up to universal constants. The theorem is intended as a standard reference whenever one needs a field-level refinement of classical generic chaining.

We also isolate a finite-dimensional pointwise-complexity principle for renormalization maps. This part is deliberately finite-cutoff: it is not a construction of a continuum interacting quantum field. Its purpose is to make precise the structural analogy with nonlinear feature learning in deep networks. An exact finite-scale telescoping identity replaces one RG scale at a time; the resulting secant response Gram defines a pointwise ellipsoidal metric; and a data-independent hierarchical prior over phase labels and active subspaces splits the complexity into a local effective dimension and a global atlas cost. We also discuss the limitations of this perspective in relation to marginal triviality, universal-approximation ansatzes for RG remainders, and the Yang–Mills mass-gap program. These connections should be viewed as possible directions of the present framework, since each would require substantial additional, model-specific estimates controlling connected correlations, RG contraction, or gauge-invariant exponential decay.

Finally, we apply the Gaussian theorem to finite pinned Gaussian free fields. Through the second Ray–Knight isomorphism, the pointwise Gaussian envelope transfers to high-probability envelopes for random-walk local times at inverse-local-time stopping rules, yielding explicit cover-time and blanket-time threshold criteria.

Contents

1	Introduction and Main Results	2
1.1	Main pointwise Gaussian theorem	2
1.2	Main finite-scale renormalization theorem	5
1.3	Organization	6
2	Proof of the Pointwise Gaussian Theorem	6
3	Finite-Scale Renormalization and Pointwise Dimension	9
3.1	Exact finite-scale telescoping for nonlinear RG maps	10
3.2	Hierarchical phase/subspace prior	12
3.3	Free Gaussian RG as the fixed-kernel baseline	13

3.4	A finite nonconvex phase-atlas example	13
3.5	Relation to ADC, universal approximation, and Yang–Mills	15
4	Graph and Local-Time Applications	16
4.1	Random walk on graphs and Gaussian free field	16
4.2	Ray–Knight bridge and high-probability analysis	17
4.3	Classical cover-time theory and relation to the present result	19
5	Scope and Outlook	20

1 Introduction and Main Results

Classical generic chaining controls the scalar quantity $\mathbb{E} \sup_{x \in T} X_x$ for a Gaussian process $\{X_x\}_{x \in T}$. For many field-level questions, however, this scalar is not the first object one wants to estimate. In the graph setting, the second Ray–Knight theorem identifies a local-time field at an inverse-local-time stopping rule with a shifted square of a Gaussian free field. In constructive and lattice field theory, one similarly studies local observables before passing to global maxima or continuum limits. This suggests that the Gaussian input should itself be pointwise: one should first prove a simultaneous field-level envelope and only afterwards take suprema or threshold events.

A further motivation comes from recent work on pointwise generalization for deep neural networks (Li and Xu, 2026). There the central object is not a class-wide supremum but a hypothesis-dependent finite-scale complexity. The proof is structural: a pointwise envelope for empirical processes bound using a mixed empirical ghost metric, an exact non-perturbative telescoping expansion exposes learned feature Gram matrices, and a hierarchical local-chart/global-atlas prior converts the resulting data-dependent active subspaces into a data-independent bound. The present paper develops the Gaussian-process counterpart of that viewpoint. Theorem 1.1 can serve as a standard reference whenever one needs a field-level refinement of classical generic chaining.

The paper then records a finite-cutoff renormalization analogue of this pointwise viewpoint. The analogy with deep feature learning is not merely that layers resemble scales. The shared proof architecture is pointwise Gaussian processes bound, exact finite-scale telescoping, pointwise ellipsoidal metric induced by learned or renormalized response Gram, and hierarchical atlas separating local effective dimension from global prior-independence cost. We state this part as an abstract deterministic complexity theorem, together with a chartwise nonconvex example. We also keep the limitations explicit: pointwise complexity can organize finite-cutoff RG geometry, but it does not by itself prove Wick factorization, marginal triviality, a universal-approximation-based RG proof, or the Yang–Mills mass gap. Those problems require additional model-specific contraction, Gaussianization, or gauge-invariant correlation estimates.

1.1 Main pointwise Gaussian theorem

Let $\{X_x\}_{x \in T}$ be a centered Gaussian process on a finite index set T . Fix an anchor $x_0 \in T$ such that $X_{x_0} = 0$ almost surely, and define

$$d(x, y) := (\mathbb{E}|X_x - X_y|^2)^{1/2}, \quad \sigma(x) := (\mathbb{E}X_x^2)^{1/2} = d(x, x_0), \quad R_* := \sup_{x \in T} \sigma(x).$$

For an ambient prior $\mu \in \Delta(T)$, where $\Delta(T)$ denotes the set of probability measures on T , define the localized pointwise Fernique–Talagrand functional

$$\Phi_\mu(x) := \int_0^{4\sigma(x)} \sqrt{\log \frac{1}{\mu(B_d(x, \varepsilon))}} d\varepsilon, \quad B_d(x, \varepsilon) := \{y \in T : d(x, y) \leq \varepsilon\}. \quad (1.1)$$

Here the upper limit is local: it is proportional to the standard deviation $\sigma(x)$, rather than the global radius. The factor 4 is inessential; it absorbs the two constant-factor losses coming from anchoring and from the nearest-point pushforward used in the ambient-equivalence step. If one instead works with the non-local full-radius integral, then the upper limits R_* and $2R_*$ are equivalent up to universal constants, since the entropy integrand is nonincreasing in ε .

Write

$$\Phi_{\mu,*} := \sup_{x \in T} \Phi_\mu(x), \quad \log_2^+(u) := \max\{\log_2 u, 0\}.$$

The degenerate case $R_* = 0$ is trivial, and the statements below are understood to be vacuous if $\Phi_{\mu,*} = +\infty$.

For $r_0 > 0$, $s_0 \in (0, R_*]$, and $\delta \in (0, 1)$, set

$$M_{r_0, s_0} := (1 + \lceil \log_2^+(\Phi_{\mu,*}/r_0) \rceil)(1 + \lceil \log_2(R_*/s_0) \rceil), \quad (1.2)$$

and

$$\mathcal{E}_{\mu, r_0, s_0}(x; \delta) := \max\{\Phi_\mu(x), r_0\} + \max\{\sigma(x), s_0\} \sqrt{\log \left(\frac{eM_{r_0, s_0}}{\delta} \right)}. \quad (1.3)$$

Thus all logarithmic losses from the two successive pointwise-peeling steps are isolated in M_{r_0, s_0} .

We state the main result as a variance-aware pointwise Gaussian majorizing-measure theorem in high probability.

Theorem 1.1 (Pointwise Gaussian majorizing-measure bound). *Assume $0 < R_* < \infty$ and $\Phi_{\mu,*} < \infty$. There exists an absolute constant $C > 0$ such that, for every ambient prior $\mu \in \Delta(T)$, every $r_0 > 0$, every $s_0 \in (0, R_*]$, and every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\forall x \in T : \quad X_x \leq C \mathcal{E}_{\mu, r_0, s_0}(x; \delta). \quad (1.4)$$

Consequently, applying the same bound to X and $-X$ and taking a union bound, with probability at least $1 - \delta$,

$$\forall x \in T : \quad |X_x| \leq A_{\mu, r_0, s_0}(x; \delta), \quad A_{\mu, r_0, s_0}(x; \delta) := C \mathcal{E}_{\mu, r_0, s_0}(x; \delta/2). \quad (1.5)$$

When r_0, s_0 are fixed, we abbreviate A_{μ, r_0, s_0} to A_μ .

Before presenting the in-expectation consequence, we record the corresponding anchored form of the classical generic-chaining scale. This is a localized reformulation of the usual Fernique–Talagrand theorem: the cutoff $4\sigma(x)$ is not the standard full-radius presentation, and therefore the global radius R_* must remain as part of the anchored scale.

Proposition 1.2 (Classical scale in anchored local form). *There exist absolute constants $0 < c < C < \infty$ such that*

$$c \left(R_* + \inf_{\mu \in \Delta(T)} \Phi_{\mu,*} \right) \leq \mathbb{E} \sup_{x \in T} X_x \leq C \left(R_* + \inf_{\mu \in \Delta(T)} \Phi_{\mu,*} \right). \quad (1.6)$$

Moreover, since the process is centered and anchored,

$$\mathbb{E} \sup_{x \in T} |X_x| \asymp \mathbb{E} \sup_{x \in T} X_x.$$

Therefore

$$\mathbb{E} \sup_{x \in T} |X_x| \asymp R_* + \inf_{\mu \in \Delta(T)} \Phi_{\mu,*}. \quad (1.7)$$

We now record the in-expectation consequence of Theorem 1.1. In particular, after optimizing over the ambient prior, the resulting global in-expectation bound is sharp up to universal constants. The variance-radius term is not an additional loss: it is part of the anchored generic-chaining scale in (1.7), while the peeling logarithms disappear by taking $r_0 \geq \Phi_{\mu,*}$ and $s_0 = R_*$.

Corollary 1.3 (In-expectation consequence and sharp global scale). *Under the assumptions of Theorem 1.1, there exists an absolute constant $C > 0$ such that*

$$\mathbb{E} \left[\sup_{x \in T} \frac{(|X_x| - C \max\{\Phi_\mu(x), r_0\})_+}{\max\{\sigma(x), s_0\}} \right] \leq C \sqrt{\log(eM_{r_0, s_0})}. \quad (1.8)$$

Consequently, for every $x \in T$,

$$\mathbb{E}|X_x| \leq C \max\{\Phi_\mu(x), r_0\} + C \max\{\sigma(x), s_0\} \sqrt{\log(eM_{r_0, s_0})}. \quad (1.9)$$

At the global level, taking $r_0 \geq \Phi_{\mu,*}$ and $s_0 = R_*$ gives $M_{r_0, s_0} = 1$, hence

$$\mathbb{E} \sup_{x \in T} |X_x| \leq C(r_0 + R_*). \quad (1.10)$$

Letting $r_0 \downarrow \Phi_{\mu,*}$, then optimizing over μ , and using (1.7), we obtain the sharp equivalence

$$\mathbb{E} \sup_{x \in T} |X_x| \asymp R_* + \inf_{\mu \in \Delta(T)} \Phi_{\mu,*}. \quad (1.11)$$

Thus the optimized expectation version is optimal up to absolute constants.

Remark (variance term versus chaining complexity). The term involving $\sigma(x)$ is a variance-sensitive Gaussian tail term. It is necessary for high-probability control, already for a single non-anchor point $X_x \sim N(0, \sigma(x)^2)$, but it is not a replacement for the chaining complexity $\Phi_\mu(x)$, which carries the local geometry of the index set.

For a fixed marginal,

$$\mathbb{E}|X_x| = \sqrt{\frac{2}{\pi}} \sigma(x),$$

so the extra factor $\sqrt{\log(eM_{r_0, s_0})}$ in (1.9) is not intrinsic to $\mathbb{E}|X_x|$. It is the price of extracting a simultaneous pointwise envelope by peeling first over the local Fernique–Talagrand complexity and then over the local variance. At the global in-expectation level, this peeling factor disappears, yielding the sharp scale in (1.11).

Remark (pointwise high probability). For Gaussian processes, subset-level in-expectation bounds transfer to high-probability bounds through Borell–Tsirelson concentration, with variance scale $\sigma_H = \sup_{x \in H} (\mathbb{E} X_x^2)^{1/2}$ for a fixed subset H . The genuinely pointwise statement is not obtained from this transfer alone: after the subset-homogeneous estimate is established, one first applies uniform pointwise peeling to localize the Fernique–Talagrand term $\Phi_\mu(x)$, and then applies the same peeling mechanism a second time to improve the tail scale from the global radius to the local variance $\sigma(x)$. Conversely, the resulting high-probability envelope implies the stated in-expectation consequence by integrating the tail.

1.2 Main finite-scale renormalization theorem

We next state the finite-dimensional renormalization theorem in the form used later. Let a depth- J RG trajectory be parametrized by scale variables

$$g = (g_0, \dots, g_{J-1}), \quad g_j \in B_2(R_j) \subset \mathbb{R}^{p_j},$$

and let $S_J(g)$ denote the final effective action, or any finite-dimensional observable extracted from the final effective action, equipped with a semimetric ρ . The theorem is conditional on a finite-scale domination statement of the form

$$\rho(S_J(g'), S_J(g))^2 \leq \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g, \varepsilon) (g'_j - g_j) \quad (1.12)$$

for all g' in a finite ρ -neighborhood of g . This defines the pointwise ellipsoidal RG metric

$$\rho_{G,g,\varepsilon}(g, g')^2 := \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g, \varepsilon) (g'_j - g_j),$$

which is the RG analogue of the pointwise ellipsoidal metric induced by learned feature Gram matrices in the DNN theory. The matrices $G_j(g, \varepsilon)$ are not infinitesimal Hessians; they are finite-scale secant response Grams produced by replacing one RG scale at a time. Section 3 proves (1.12) from an exact telescoping identity whenever the one-step RG replacements have secant covariance bounds and the later RG flow is locally stable.

For a positive semidefinite matrix G , radius R , and resolution $u > 0$, define

$$r_{\text{eff}}(G, R, u) := \max\{k : \lambda_k(G)R^2 \geq u^2\}, \quad d_{\text{ell}}(G, R, u) := \frac{1}{2} \sum_{k=1}^{r_{\text{eff}}} \log \left(\frac{CR^2 \lambda_k(G)}{u^2} \right), \quad (1.13)$$

with eigenvalues in nonincreasing order. Let $V_j(g, \varepsilon)$ be the active eigenspace of $G_j(g, \varepsilon)$, and let $\mathfrak{A}_j(g, \varepsilon, u_j)$ be the atlas cost of choosing the phase label, effective rank, and a reference subspace close to $V_j(g, \varepsilon)$. A concrete Grassmannian expression is given in Section 3.

Theorem 1.4 (Finite-scale RG pointwise dimension, main form). *Assume the finite-scale domination (1.12). Choose numbers $u_j > 0$ such that $\sum_j u_j^2 \leq c\varepsilon^2$. Then there exists a data-independent hierarchical prior Π over phase labels, active ranks, reference subspaces, and local scale couplings such that*

$$\log \frac{1}{\Pi(B_\rho(g, \varepsilon))} \leq C \sum_{j=0}^{J-1} [d_{\text{ell}}(G_j(g, \varepsilon), R_j, u_j) + \mathfrak{A}_j(g, \varepsilon, u_j)]. \quad (1.14)$$

Thus finite-scale renormalization has the same complexity structure as nonlinear feature-learning DNN: a local ellipsoidal effective dimension plus a global atlas cost at each scale.

The theorem should be read as an abstract finite-cutoff theorem, not as a shortcut to continuum constructive QFT. To prove a scaling-limit theorem such as four-dimensional marginal triviality, one also needs a model-specific contraction or Gaussianization estimate for non-Gaussian connected correlations. The pointwise-dimension theorem supplies the local finite-scale complexity layer once such RG or correlation estimates are available.

1.3 Organization

Section 2 proves the pointwise Gaussian theorem. Section 3 develops the finite-scale renormalization pointwise-complexity theory, including the DNN–RG dictionary, secant-Gram metric domination, phase/subspace atlas covering, and the precise relation to marginal triviality, universal approximation, and Yang–Mills. Section 4 gives the graph and local-time applications, including the Ray–Knight corollaries and the relation to classical cover-time theory. Section 5 summarizes the finite-cutoff scope and the main lessons.

2 Proof of the Pointwise Gaussian Theorem

The proof follows a subset-homogeneity and peeling strategy. First, for each fixed subset $H \subseteq T$, we combine an anchored form of Fernique’s majorizing-measure upper bound (Fernique, 1974; Talagrand, 1987), the ambient equivalence principle of pointwise dimension, and Borell–Tsirelson concentration. Second, we apply the one-parameter uniform pointwise peeling lemma twice: first to localize the Fernique–Talagrand integral $\Phi_\mu(x)$, and then to localize the variance scale from R_* to $\sigma(x)$. This avoids introducing a separate two-dimensional peeling corollary.

Lemma 2.1 (Uniform pointwise peeling for indexed processes). *Let $\{Z_x\}_{x \in T}$ be random variables, let $d_0 : T \rightarrow [0, R]$, and let $\psi(r; \delta)$ be nondecreasing in r and nonincreasing in δ . Assume that for every fixed subset $H \subseteq T$ and every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(\sup_{x \in H} Z_x \leq \sup_{x \in H} \psi(d_0(x); \delta)\right) \geq 1 - \delta. \quad (2.1)$$

Then for every $r_0 > 0$, with probability at least $1 - \delta$, uniformly over all $x \in T$,

$$Z_x \leq \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{1 + \lceil \log_2^+(R/r_0) \rceil}\right), \quad (2.2)$$

where $\log_2^+(u) = \max\{\log_2 u, 0\}$.

Proof. Let $m := \lceil \log_2^+(R/r_0) \rceil$. The sets

$$H_0 := \{x : d_0(x) \leq r_0\}, \quad H_j := \{x : 2^{j-1}r_0 < d_0(x) \leq 2^j r_0\} \quad (1 \leq j \leq m)$$

cover T . Apply (2.1) to each H_j with confidence level $\delta/(m+1)$, and take a union bound. On the resulting event, if $x \in H_0$, then

$$Z_x \leq \psi\left(r_0; \frac{\delta}{m+1}\right) \leq \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{m+1}\right),$$

and if $x \in H_j$ for $j \geq 1$, then $2^j r_0 \leq 2d_0(x)$, so

$$Z_x \leq \psi\left(2^j r_0; \frac{\delta}{m+1}\right) \leq \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{m+1}\right).$$

This proves the claim. $\square$

We also use the following localized version of the ambient equivalence principle of pointwise dimension from Lemma 7 of Li and Xu (2026).

Lemma 2.2 (Ambient equivalence for the localized integral). *Let (T, d) be a finite metric space and let $H \subseteq T$. Let $p : T \rightarrow H$ be a nearest-point selector, so that $d(y, p(y)) = \min_{h \in H} d(y, h)$, and let $\mu_H = p_{\#}\mu$ be the pushforward of an ambient prior $\mu \in \Delta(T)$. Then, for every $x \in H$ and every $\varepsilon > 0$,*

$$\mu_H(B_d(x, 2\varepsilon)) \geq \mu(B_d(x, \varepsilon)).$$

Consequently, for the localized functional Φ in (1.1),

$$\Phi_{\mu_H}(x) \leq 2\Phi_{\mu}(x), \quad x \in H. \quad (2.3)$$

Proof. If $y \in B_d(x, \varepsilon)$, then

$$d(p(y), x) \leq d(p(y), y) + d(y, x) \leq 2d(y, x) \leq 2\varepsilon,$$

because $x \in H$ and $p(y)$ is a nearest point in H . Hence $p(B_d(x, \varepsilon)) \subseteq B_d(x, 2\varepsilon)$, which gives the ball-mass inequality. Therefore

$$\Phi_{\mu_H}(x) = \int_0^{4\sigma(x)} \sqrt{\log \frac{1}{\mu_H(B_d(x, u))}} du \leq 2 \int_0^{2\sigma(x)} \sqrt{\log \frac{1}{\mu(B_d(x, \varepsilon))}} d\varepsilon \leq 2\Phi_{\mu}(x),$$

where we used the change of variables $u = 2\varepsilon$. $\square$

Lemma 2.3 (Anchored subset majorizing-measure bound). *For every fixed subset $H \subseteq T$, every ambient prior $\mu \in \Delta(T)$, and*

$$\sigma_H := \sup_{x \in H} \sigma(x),$$

there is an absolute constant $C > 0$ such that

$$\mathbb{E} \sup_{x \in H} X_x \leq C \sup_{x \in H} \Phi_{\mu}(x) + C\sigma_H. \quad (2.4)$$

Proof. Let $H_0 := H \cup \{x_0\}$, and let $p : T \rightarrow H_0$ be a nearest-point selector. Define the probability measure

$$\bar{\mu}_H := \frac{1}{2}\delta_{x_0} + \frac{1}{2}p_{\#}\mu$$

on H_0 . Fernique's majorizing-measure upper bound (Fernique, 1974; Talagrand, 1987), applied to the restricted process on H_0 , gives

$$\mathbb{E} \sup_{x \in H} X_x \leq C \sup_{x \in H_0} \int_0^{2\sigma_H} \sqrt{\log \frac{1}{\bar{\mu}_H(B_d(x, u))}} du,$$

since $\text{diam}(H_0, d) \leq 2\sigma_H$. For $x = x_0$, the integral is at most $2\sigma_H\sqrt{\log 2}$, because $\bar{\mu}_H\{x_0\} \geq 1/2$. For $x \in H$, split the integral at $4\sigma(x)$. On $[4\sigma(x), 2\sigma_H]$, the ball $B_d(x, u)$ contains x_0 , hence has $\bar{\mu}_H$ -mass at least $1/2$, so this part is bounded by $C\sigma_H$. On $[0, 4\sigma(x)]$, Lemma 2.2 and the extra factor $1/2$ in $\bar{\mu}_H$ give a bound of order $\Phi_{\mu}(x) + \sigma(x)$. Taking the supremum over $x \in H$ proves (2.4). $\square$

Lemma 2.4 (Subset-homogeneous Gaussian estimate). *For every fixed subset $H \subseteq T$, every ambient prior $\mu \in \Delta(T)$, and every $\delta \in (0, 1)$,*

$$\mathbb{P} \left(\sup_{x \in H} X_x \leq C \sup_{x \in H} \Phi_{\mu}(x) + C\sigma_H \sqrt{\log(e/\delta)} \right) \geq 1 - \delta, \quad (2.5)$$

where $\sigma_H = \sup_{x \in H} \sigma(x)$.

Proof. By Lemma 2.3,

$$\mathbb{E} \sup_{x \in H} X_x \leq C \sup_{x \in H} \Phi_\mu(x) + C\sigma_H.$$

Realize $X_x = \langle g, v_x \rangle$ as an isonormal process on a Hilbert space. The map $g \mapsto \sup_{x \in H} \langle g, v_x \rangle$ is σ_H -Lipschitz. Borell–Tsirelson concentration therefore gives the desired inequality, after absorbing the expectation term $C\sigma_H$ into $C\sigma_H \sqrt{\log(e/\delta)}$. $\square$

Proof of Theorem 1.1. Fix an arbitrary subset $H \subseteq T$. Applying Lemma 2.1 on the index set H , with $Z_x = X_x$, $d_0(x) = \Phi_\mu(x)$, $R = \Phi_{\mu,*}$, and using the subset estimate (2.5), yields the following: for every $r_0 > 0$ and every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\forall x \in H : \quad X_x \leq C \max\{\Phi_\mu(x), r_0\} + C\sigma_H \sqrt{\log\left(\frac{eM_\Phi}{\delta}\right)}, \quad (2.6)$$

where

$$M_\Phi := 1 + \lceil \log_2^+(\Phi_{\mu,*}/r_0) \rceil.$$

Since H was arbitrary, (2.6) is now a subset-homogeneous estimate for the centered residual

$$Y_x := X_x - C \max\{\Phi_\mu(x), r_0\}$$

with complexity $\sigma(x)$. Applying Lemma 2.1 a second time, now with $Z_x = Y_x$, $d_0(x) = \sigma(x)$, $R = R_*$, and

$$\psi(s; \delta) := Cs \sqrt{\log\left(\frac{eM_\Phi}{\delta}\right)},$$

gives, with probability at least $1 - \delta$,

$$\forall x \in T : \quad Y_x \leq C \max\{\sigma(x), s_0\} \sqrt{\log\left(\frac{eM_\Phi M_\sigma}{\delta}\right)},$$

where

$$M_\sigma := 1 + \lceil \log_2(R_*/s_0) \rceil.$$

Combining the last display with the definition of Y_x proves (1.4), since $M_{r_0, s_0} = M_\Phi M_\sigma$. The absolute-value estimate (1.5) follows by applying the same argument to $-X$ and taking a union bound. $\square$

Proof of Proposition 1.2. The upper bound follows from Lemma 2.3 with $H = T$, followed by optimization over μ . For the lower bound, choose x_* with $\sigma(x_*) = R_*$. Since $X_{x_0} = 0$,

$$\sup_{x \in T} X_x \geq (X_{x_*})_+, \quad \mathbb{E}(X_{x_*})_+ = \frac{R_*}{\sqrt{2\pi}}.$$

It remains to lower-bound the optimized local functional. Let

$$\Phi_\mu^{\text{full}}(x) := \int_0^{2R_*} \sqrt{\log \frac{1}{\mu(B_d(x, \varepsilon))}} d\varepsilon.$$

The classical majorizing-measure theorem gives

$$\inf_{\mu \in \Delta(T)} \sup_x \Phi_\mu^{\text{full}}(x) \lesssim \mathbb{E} \sup_{x \in T} X_x.$$

Since $\Phi_\mu(x) \leq \Phi_\mu^{\text{full}}(x)$ for every x , we have

$$\inf_{\mu} \Phi_{\mu,*} \leq \inf_{\mu} \sup_x \Phi_\mu^{\text{full}}(x) \lesssim \mathbb{E} \sup_{x \in T} X_x.$$

Together with the radius lower bound this proves the left side of (1.6). The absolute-value statement follows from

$$\sup_x X_x \leq \sup_x |X_x| \leq \sup_x X_x + \sup_x (-X_x)$$

and symmetry. $\square$

Proof of Corollary 1.3. Let

$$Y := \sup_{x \in T} \frac{(|X_x| - C \max\{\Phi_\mu(x), r_0\})_+}{\max\{\sigma(x), s_0\}}.$$

By the two-sided form of Theorem 1.1, after increasing C if necessary,

$$\mathbb{P} \left(Y > C \sqrt{\log \left(\frac{eM_{r_0, s_0}}{\delta} \right)} \right) \leq \delta, \quad \delta \in (0, 1).$$

Integrating this tail gives

$$\mathbb{E}Y \leq C \sqrt{\log(eM_{r_0, s_0})},$$

which proves (1.8). The pointwise bound (1.9) follows by dropping the supremum, and the global bound (1.10) follows by taking $r_0 \geq \Phi_{\mu,*}$, $s_0 = R_*$, and then letting $r_0 \downarrow \Phi_{\mu,*}$. Optimizing over μ and applying Proposition 1.2 proves (1.11). $\square$

3 Finite-Scale Renormalization and Pointwise Dimension

This section makes precise the analogy with nonlinear feature learning. The relevant correspondence is not the superficial statement that layers are scales. The correspondence is a proof mechanism: exact finite-scale telescoping, a learned or renormalized secant Gram matrix, and a hierarchical prior over local charts and a global atlas. This is the mechanism that makes the DNN pointwise theory computable, and it is also the mechanism that a finite-cutoff nonlinear RG analysis can use. The language is compatible with rigorous Wilsonian RG frameworks such as [Brydges and Slade \(2015\)](#); [Bauerschmidt et al. \(2019\)](#), but the theorem below is an abstract finite-dimensional statement.

Deep-network pointwise theory	Finite-scale RG analogue
Layer index ℓ	RG scale index j
Weights W_ℓ	Scale coupling vector g_j
Feature matrix $F_{\ell-1}(W, X)$	Renormalized operator family $O_j(S_j)$
Feature Gram $F_{\ell-1} F_{\ell-1}^\top$	Secant response Gram Γ_j^{sec}
Exact layer telescoping	Exact replacement over RG scales
Outer Lipschitz $M_{\ell \rightarrow L}$	Stability of later RG maps $M_{j \rightarrow J}$
Active feature eigenspace	Relevant/marginal active operator subspace
Grassmannian prior	Phase/subspace atlas prior
Feature-rank compression	Contraction or truncation of irrelevant directions
Kernel/NTK baseline	Free or massive GFF baseline

All statements below are finite-dimensional and finite-cutoff. They are intended as a rigorous structural layer, not as a continuum construction of singular interacting fields.

3.1 Exact finite-scale telescoping for nonlinear RG maps

Let $\mathcal{S}_0, \dots, \mathcal{S}_J$ be finite-dimensional normed spaces. For each scale j , let

$$\mathcal{R}_j^{g_j} : \mathcal{S}_j \rightarrow \mathcal{S}_{j+1}, \quad g_j \in B_2(R_j) \subset \mathbb{R}^{p_j},$$

be a nonlinear coarse-graining map. Starting from a fixed $S_0 \in \mathcal{S}_0$, define the RG trajectory

$$S_{j+1}(g) := \mathcal{R}_j^{g_j}(S_j(g)), \quad S_J(g) = \mathcal{R}_{J-1}^{g_{J-1}} \circ \dots \circ \mathcal{R}_0^{g_0}(S_0).$$

This notation covers, for example, the classical Wilsonian integration identity

$$e^{-S_{j+1}(\Phi)} = \int e^{-S_j(\Phi + \psi_j)} d\gamma_j(\psi_j).$$

This is the finite-dimensional analogue of integrating out one fluctuation scale. Perturbative RG expands this map; here we keep the finite map itself.

For two coupling sequences g, g' , define the hybrid trajectory

$$S_j^{(g', g; m)} := \mathcal{R}_{j-1}^{g'_{j-1}} \circ \dots \circ \mathcal{R}_m^{g'_m} \circ \mathcal{R}_{m-1}^{g_{m-1}} \circ \dots \circ \mathcal{R}_0^{g_0}(S_0),$$

with the evident convention at the endpoints. The exact formula below is an elementary finite-scale telescoping identity for compositions of nonlinear maps, used here to parallel the non-perturbative layerwise expansion in [Li and Xu \(2026\)](#).

Lemma 3.1 (Exact RG replacement telescoping). *For every g, g' ,*

$$S_J(g') - S_J(g) = \sum_{j=0}^{J-1} \left[\mathcal{R}_{j-1:j+1}^{g'}(\mathcal{R}_j^{g'_j}(S_j^{(j)})) - \mathcal{R}_{j-1:j+1}^{g'}(\mathcal{R}_j^{g_j}(S_j^{(j)})) \right], \quad (3.1)$$

where $S_j^{(j)} := \mathcal{R}_{j-1}^{g_{j-1}} \circ \dots \circ \mathcal{R}_0^{g_0}(S_0)$ and $\mathcal{R}_{j-1:j+1}^{g'} := \mathcal{R}_{j-1}^{g'_{j-1}} \circ \dots \circ \mathcal{R}_{j+1}^{g'_{j+1}}$.

Proof. Insert the hybrid trajectories that use g up to scale $j-1$ and g' from scale j onward. The difference between two consecutive hybrids changes only the scale- j map. Summing these consecutive differences gives (3.1). $\square$

Let ρ be a semimetric on $\mathcal{S}_J$. The next result is the RG analogue of the non-perturbative DNN feature expansion: the finite replacement identity induces an ellipsoidal metric whose matrices are finite-scale secant response Grams.

Theorem 3.2 (Secant-Gram metric domination for nonlinear RG). *Fix g and a finite neighborhood $\mathcal{N}(g, \varepsilon)$. Assume that for each j :*

(i) *the later RG flow is locally stable: for all states U, V arising in $\mathcal{N}(g, \varepsilon)$,*

$$\rho(\mathcal{R}_{j-1:j+1}^{g'}(U), \mathcal{R}_{j-1:j+1}^{g'}(V)) \leq M_{j \rightarrow J}(g, \varepsilon) \|U - V\|_{j+1};$$

(ii) *the scale- j finite replacement is dominated by a positive semidefinite secant Gram $\Gamma_j(g, \varepsilon)$:*

$$\|\mathcal{R}_j^{g'_j}(S_j^{(j)}) - \mathcal{R}_j^{g_j}(S_j^{(j)})\|_{j+1}^2 \leq (g'_j - g_j)^\top \Gamma_j(g, \varepsilon) (g'_j - g_j).$$

Then all $g' \in \mathcal{N}(g, \varepsilon)$ satisfy

$$\rho(S_J(g'), S_J(g))^2 \leq \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g, \varepsilon) (g'_j - g_j), \quad G_j := J M_{j \rightarrow J}(g, \varepsilon)^2 \Gamma_j(g, \varepsilon). \quad (3.2)$$

Proof. Apply Lemma 3.1, the triangle inequality for ρ , the local stability assumption, and the one-step secant-Gram bound. Cauchy-Schwarz gives a factor J when the J increments are summed; this factor is included in the definition of G_j . $\square$

How the secant Gram is verified. Condition (ii) in Theorem 3.2 is a finite-dimensional secant calculation. Fix the state $S_j^{(j)}$ and set

$$F_j(g_j) := \mathcal{R}_j^{g_j}(S_j^{(j)}).$$

If F_j is continuously Frechet differentiable on the finite neighborhood under consideration and the next-scale norm is Hilbertian, then for $\Delta_j = g'_j - g_j$ and $g_{j,t} = g_j + t\Delta_j$,

$$F_j(g'_j) - F_j(g_j) = \int_0^1 DF_j(g_{j,t})[\Delta_j]dt,$$

and Jensen's inequality gives

$$\|F_j(g'_j) - F_j(g_j)\|_{j+1}^2 \leq \Delta_j^\top \left(\int_0^1 DF_j(g_{j,t})^* DF_j(g_{j,t}) dt \right) \Delta_j. \quad (3.3)$$

Thus condition (ii) holds with the pairwise secant Gram in parentheses, or with any positive-semidefinite Loewner envelope that dominates it uniformly over the finite neighborhood. For an exponential-family RG step

$$\mathcal{R}_j^g(S)(\Phi) = -\log \int \exp \left\{ -S(\Phi + \psi) - \sum_{a=1}^{p_j} g_a O_{j,a}(\Phi + \psi) \right\} d\gamma_j(\psi),$$

one has

$$\partial_{g_a} \mathcal{R}_j^g(S)(\Phi) = \mathbb{E}_{j,\Phi,g}[O_{j,a}(\Phi + \psi)].$$

Consequently the secant Gram is the Gram matrix of the finite-scale response functions

$$\Gamma_{j,ab}^{\text{sec}}(g, g') = \int_0^1 \langle m_{j,a}^t, m_{j,b}^t \rangle_{j+1} dt, \quad m_{j,a}^t(\Phi) := \mathbb{E}_{j,\Phi,g,t}[O_{j,a}(\Phi + \psi)].$$

This is the precise RG analogue of the learned feature Gram matrix in the DNN pointwise metric.

Compressed response Grams. The scalar Lipschitz choice $\Gamma_j = L_j^2 I$ verifies condition (ii), but it gives no pointwise gain. A useful pointwise-complexity theorem requires an anisotropic response estimate. Let $K_{j+1 \rightarrow J}$ be a positive-semidefinite quadratic form measuring the sensitivity of the later RG flow from scale $j+1$ to the terminal observable. The propagated response Gram is then

$$G_j^{\text{RG}}(g, g') := \int_0^1 DF_j(g_{j,t})^* K_{j+1 \rightarrow J} DF_j(g_{j,t}) dt. \quad (3.4)$$

Suppose that the coupling space decomposes as $\mathbb{R}^{p_j} = U_j \oplus U_j^\perp$. Let P_j be an orthonormal basis matrix for U_j . Assume that, uniformly over the finite neighborhood,

$$\|DF_j(\theta)P_j a\|_{K_{j+1 \rightarrow J}}^2 \leq a^\top A_j a \quad (a \in \mathbb{R}^{\dim U_j}), \quad \|DF_j(\theta)v\|_{K_{j+1 \rightarrow J}}^2 \leq \tau_j \|v\|^2 \quad (v \in U_j^\perp).$$

Then, for every g' in the neighborhood,

$$G_j^{\text{RG}}(g, g') \preceq 2P_j A_j P_j^\top + 2\tau_j I. \quad (3.5)$$

Indeed, write any direction $h \in \mathbb{R}^{p_j}$ as $h = P_j a + v$, with $v \in U_j^\perp$, and use $\|\xi + \zeta\|^2 \leq 2\|\xi\|^2 + 2\|\zeta\|^2$. Integrating along the secant path gives (3.5) in the Loewner order. After diagonalizing A_j , the active eigenvalues give the local ellipsoidal effective dimension, while directions with $\tau_j R_j^2 \lesssim u_j^2$ are below finite resolution and do not contribute. In perturbative RG language, U_j is the finite-resolution span of relevant and marginal operator directions, while irrelevant operators contribute only the residual term. In a nonconvex model this compression is imposed chartwise, and the phase/subspace prior below pays the cost of selecting the chart and active subspace.

3.2 Hierarchical phase/subspace prior

For a positive semidefinite matrix $G \in \mathbb{R}^{p \times p}$, a Euclidean radius R , and a resolution $u > 0$, set

$$r_{\text{eff}}(G, R, u) := \max\{k : \lambda_k(G)R^2 \geq u^2\}, \quad V_{\text{eff}}(G, R, u) := \text{span}\{\text{top } r_{\text{eff}} \text{ eigenvectors}\},$$

and

$$d_{\text{ell}}(G, R, u) := \frac{1}{2} \sum_{k=1}^{r_{\text{eff}}(G, R, u)} \log \left(\frac{CR^2 \lambda_k(G)}{u^2} \right). \quad (3.6)$$

The following elementary estimate is the local-chart calculation used throughout. For subspaces $V, \bar{V} \subset \mathbb{R}^p$, write

$$\rho_{\text{proj}, G}(V, \bar{V}) := \|G^{1/2}(P_V - P_{\bar{V}})\|_{\text{op}}.$$

Lemma 3.3 (Ellipsoidal local chart). *Let $r = r_{\text{eff}}(G, R, u)$, let $V = V_{\text{eff}}(G, R, u)$, and let $\bar{V} \in \text{Gr}(p, r)$ satisfy*

$$\rho_{\text{proj}, G}(V, \bar{V}) \leq \frac{u}{4R}.$$

Let $\pi_{\bar{V}}$ be the uniform law on $B_2(2R) \cap \bar{V}$. Then, after changing the universal numerical constant in the radius, uniformly over $g \in B_2(R)$,

$$\log \frac{1}{\pi_{\bar{V}}\{g' : (g' - g)^\top G(g' - g) \leq Cu^2\}} \leq C + d_{\text{ell}}(G, R, u).$$

Equivalently, replacing u by a constant multiple gives the displayed local-chart bound used in Theorem 3.4.

Proof. The point is that the prior may be supported on the reference subspace $\bar{V}$ even though the center g need not belong to $\bar{V}$. Since $r = r_{\text{eff}}(G, R, u)$, the component of any vector in the inactive subspace $V^\perp$ has G -length at most u on the Euclidean ball $B_2(R)$. The projection-distance assumption transfers the active subspace V to $\bar{V}$ with additional G -error at most $u/4$. Hence the G -ball centered at g , with radius enlarged by a universal constant, contains a translate inside $\bar{V}$ of an r -dimensional ellipsoid whose principal semi-axes are comparable to

$$\min \left\{ R, \frac{u}{\sqrt{\lambda_k(G)}} \right\}, \quad k = 1, \dots, r.$$

Because the support of $\pi_{\bar{V}}$ is $B_2(2R) \cap \bar{V}$, this translated ellipsoid lies in the support up to another universal loss. The volume ratio between $B_2(2R) \cap \bar{V}$ and this ellipsoid is therefore bounded by

$$C \prod_{k=1}^r \left(\frac{CR\sqrt{\lambda_k(G)}}{u} \right),$$

with factors below one omitted precisely by the definition of r_{eff} . Taking logarithms gives $C + d_{\text{ell}}(G, R, u)$. This is the same finite-dimensional volume-ratio argument used in the DNN local-chart lemma. $\square$

Let $\mathcal{A}_j$ be a finite set of phase labels at scale j , with prior weights $q_j(a) > 0$. Conditional on a phase label a and rank r , let $\nu_{j,a,r}$ be a prior on the Grassmannian $\text{Gr}(p_j, r)$. For $G_j = G_j(g, \varepsilon)$, define the atlas cost

$$\mathfrak{A}_j(g, \varepsilon, u_j) := \log \frac{1}{q_j(a_j(g))} + \log(1 + p_j) + \log \frac{1}{\nu_{j,a_j,r_j}(B_{\rho_{\text{proj}, G_j}}(V_j, u_j/(4R_j)))}, \quad (3.7)$$

where $r_j = r_{\text{eff}}(G_j, R_j, u_j)$, $V_j = V_{\text{eff}}(G_j, R_j, u_j)$, and $a_j(g)$ is any chart containing the trajectory at scale j . If $\nu_{j,a,r}$ is Haar-uniform on $\text{Gr}(p_j, r)$, the last term is bounded by the usual Grassmannian entropy; in an ellipsoidal projection metric one may use the anisotropic Grassmannian bound from the DNN pointwise theory (Li and Xu, 2026).

Theorem 3.4 (Finite-scale RG pointwise dimension). *Assume the metric domination (3.2). Choose resolutions $u_j > 0$ such that $\sum_{j=0}^{J-1} u_j^2 \leq c\varepsilon^2$. Construct a hierarchical prior Π by independently, at each scale j , sampling a phase label a , an effective rank r , a reference subspace $\bar{V} \in \text{Gr}(p_j, r)$, and then sampling g_j uniformly in $B_2(2R_j) \cap \bar{V}$. Then, uniformly for every trajectory g ,*

$$\log \frac{1}{\Pi(B_\rho(g, \varepsilon))} \leq C \sum_{j=0}^{J-1} [d_{\text{ell}}(G_j(g, \varepsilon), R_j, u_j) + \mathfrak{A}_j(g, \varepsilon, u_j)]. \quad (3.8)$$

Proof. For each scale choose the phase label $a_j(g)$, the active rank r_j , and a reference subspace $\bar{V}_j$ within projection distance $u_j/(4R_j)$ of V_j . Lemma 3.3 gives the local prior mass of the ellipsoidal u_j -ball inside this chart. Multiplying the scale-wise prior masses and including the phase, rank, and subspace probabilities gives the right-hand side of (3.8). Finally, if each scale perturbation lies in its ellipsoidal u_j -ball, then (3.2) and $\sum_j u_j^2 \leq c\varepsilon^2$ imply $g' \in B_\rho(g, \varepsilon)$. Hence the product event is contained in the ρ -ball, and its prior mass lower-bounds $\Pi(B_\rho(g, \varepsilon))$. $\square$

Interpretation. The theorem is the RG counterpart of the DNN Riemannian-dimension calculation. The local term d_{ell} is the effective dimension of the finite-scale response Gram. The atlas term is the price of making the prior independent of the realized RG trajectory. Irrelevant directions do not contribute once their eigenvalues fall below the finite resolution u_j^2/R_j^2 ; relevant and marginal directions are precisely the active directions that remain in the sum.

3.3 Free Gaussian RG as the fixed-kernel baseline

Let $E = E_C \oplus E_F$ be finite-dimensional and let a centered Gaussian field have precision matrix

$$Q = \begin{pmatrix} Q_{CC} & Q_{CF} \\ Q_{FC} & Q_{FF} \end{pmatrix}, \quad Q_{FF} \succ 0.$$

Define the Schur-complement precision

$$Q_C^{\text{RG}} := Q_{CC} - Q_{CF} Q_{FF}^{-1} Q_{FC}.$$

Then marginalizing the fluctuation variables gives $\phi_C \sim N(0, (Q_C^{\text{RG}})^{-1})$, and conditionally

$$\phi_F = -Q_{FF}^{-1} Q_{FC} \phi_C + \zeta, \quad \zeta \sim N(0, Q_{FF}^{-1})$$

independently of ϕ_C . This is a fixed-kernel RG step: all pointwise complexity is determined by the Schur-complement covariance and the fluctuation covariance. It is therefore analogous to a kernel, GP, or NTK model rather than to nonlinear feature learning.

3.4 A finite nonconvex phase-atlas example

The previous theorem is abstract. We now give a concrete finite-volume chartwise example. The point is not to construct a full continuum double-well field, but to show how a globally non-log-concave finite measure can be controlled after it is decomposed into stable phase charts. Let

$E = E_C \oplus E_F$, let $\mathcal{A}$ be a finite set of phases, and let

$$\nu(d\phi_C, d\phi_F) = \sum_{a \in \mathcal{A}} w_a \nu_a(d\phi_C, d\phi_F), \quad \nu_a \propto e^{-S_a(\phi_C, \phi_F)} \mathbf{1}_{\Omega_{C,a} \times \Omega_{F,a}} d\phi_C d\phi_F,$$

where the sets $\Omega_{C,a}, \Omega_{F,a}$ are closed convex sets with nonempty interior. The mixture may be globally non-log-concave and multimodal. Assume that, on each chart,

$$\nabla^2 S_a(\phi_C, \phi_F) \succeq H_a = \begin{pmatrix} A_a & B_a \\ B_a^\top & D_a \end{pmatrix}, \quad D_a \succ 0, \quad H_a^{\text{RG}} := A_a - B_a D_a^{-1} B_a^\top \succ 0.$$

Theorem 3.5 (One-step nonconvex phase-atlas RG stability). *Under the preceding assumptions, the coarse marginal is again a finite phase-atlas mixture*

$$\nu_C(d\phi_C) = \sum_{a \in \mathcal{A}} w_a^+ \nu_{C,a}(d\phi_C), \quad \nu_{C,a} \propto e^{-S_a^{\text{RG}}(u)} \mathbf{1}_{\Omega_{C,a}} du,$$

where the weights w_a^+ are the corresponding normalized marginal weights and

$$S_a^{\text{RG}}(u) := -\log \int_{\Omega_{F,a}} e^{-S_a(u,v)} dv.$$

Moreover, S_a^{RG} is H_a^{RG} -strongly convex. Hence, for any finite family of coarse linear observables $Y_x(u) = \langle v_x, u \rangle$, the centered chart process $Y_x - \mathbb{E}_{\nu_{C,a}} Y_x$ is sub-Gaussian with canonical metric

$$d_a(x, y)^2 = \langle v_x - v_y, (H_a^{\text{RG}})^{-1} (v_x - v_y) \rangle.$$

Consequently, the standard sub-Gaussian chaining-and-peeling argument applied chart by chart, with the comparison metric induced by $(H_a^{\text{RG}})^{-1}$, yields a simultaneous envelope for the mixture, with an additional phase-atlas cost $\log(1/\min_a w_a^+)$, or $\log |\mathcal{A}|$ for equal weights.

Proof. The marginal formula is the definition of integration over E_F . The Schur-complement lower bound follows by completing the square:

$$\begin{pmatrix} u \\ v \end{pmatrix}^\top H_a \begin{pmatrix} u \\ v \end{pmatrix} = u^\top H_a^{\text{RG}} u + (v + D_a^{-1} B_a^\top u)^\top D_a (v + D_a^{-1} B_a^\top u).$$

Thus $S_a(u, v) - \frac{1}{2} u^\top H_a^{\text{RG}} u$ is jointly convex after the harmless affine shift in v . By the Prekopa–Leindler theorem, integrating out v preserves log-concavity, so $S_a^{\text{RG}}(u) - \frac{1}{2} u^\top H_a^{\text{RG}} u$ is convex. Brascamp–Lieb and Herbst then give the stated sub-Gaussian bound for linear observables. The envelope follows from the usual generic-chaining upper bound for sub-Gaussian increments, followed by the same pointwise peeling used in the Gaussian proof and a union bound over $a \in \mathcal{A}$. $\square$

Corollary 3.6 (Pure-phase charts for a finite-volume double-well Φ^4 model). *Let $G = (V, E)$ be finite and*

$$S(\phi) = \frac{1}{2} \sum_{\{u,v\} \in E} c_{uv} (\phi_u - \phi_v)^2 + \lambda \sum_{u \in V} (\phi_u^2 - a^2)^2.$$

Fix $m > a/\sqrt{3}$ and define phase charts

$$\Omega_+ = \{\phi : \phi_u \geq m \ \forall u\}, \quad \Omega_- = -\Omega_+.$$

The restrictions of $e^{-S(\phi)}d\phi$ to Ω_+ and Ω_- define two stable phase charts. A finite mixture of these chart measures is generally non-log-concave, but every chart has Hessian bounded below by

$$L_G + 4\lambda(3m^2 - a^2)I.$$

Therefore Theorem 3.5 applies to every finite decomposition of variables $V = C \sqcup F$, and one nonlinear coarse-graining step preserves a two-chart pointwise-complexity envelope for these pure-phase restrictions. To cover the full double-well Gibbs measure one would need additional charts, for instance droplet or interface charts, and corresponding chartwise Hessian bounds.

Proof. On either chart, $|\phi_u| \geq m$, and

$$\frac{d^2}{dt^2}\lambda(t^2 - a^2)^2 = 4\lambda(3t^2 - a^2) \geq 4\lambda(3m^2 - a^2) > 0.$$

Adding the graph Dirichlet Hessian gives the displayed lower bound. The result follows from Theorem 3.5. $\square$

3.5 Relation to ADC, universal approximation, and Yang–Mills

The finite-scale theorem above controls metric-ball complexity of an RG trajectory or of chartwise coarse observables. It does not by itself prove Wick factorization, Gaussianity of a scaling limit, or a mass gap. This distinction is essential.

Relation to Aizenman–Duminil-Copin. In the four-dimensional Ising and lattice $\lambda\Phi_4^4$ setting, Aizenman and Duminil-Copin (2021) prove marginal triviality by controlling deviations from Wick’s rule through random currents and multiscale estimates. The difficulty there is not merely to obtain an a priori amplitude bound on the field. Indeed, in the Ising case the microscopic spins are already uniformly bounded. The essential task is instead to prove Gaussianization of macroscopic observables, or equivalently to control the non-Gaussian connected-correlation remainder. Therefore, Theorem 3.4 should not be viewed as a complete route to reproving their theorem on its own. Its possible role is complementary: once a model-specific random-current or rigorous-RG argument provides Gaussianization, the pointwise-complexity theorem gives a local, finite-cutoff, and observable-sensitive envelope whose cost is carried only by active response-Gram directions and by the phase/subspace atlas.

Universal approximation is not a certificate. One might try to approximate the RG remainder by a neural network, using universal approximation theorems (Cybenko, 1989; Hornik et al., 1989). At a fixed finite cutoff this can approximate a continuous finite-dimensional RG map on a compact set. However, universal approximation alone gives neither a scale-uniform error bound nor preservation of locality, reflection positivity, correlation inequalities, or Wick-factorization errors. To obtain an ADC-type theorem through a DNN-parametrized RG ansatz, one would need certified estimates of the form

$$\|\mathcal{R}_j(S_j) - \widehat{\mathcal{R}}_j(S_j)\|_{\mathcal{N}_{j+1}} \leq \eta_j, \quad \sum_j \eta_j < \infty,$$

in a norm $\mathcal{N}_j$ that controls connected correlations, together with a marginal-flow estimate such as

$$\lambda_{j+1} \leq \lambda_j - c\lambda_j^2 + C\eta_j$$

and contraction of irrelevant remainders. These are model-specific RG estimates, not consequences of approximation theory or covering numbers alone.

Yang–Mills. For Yang–Mills, the corresponding finite-cutoff program would have to be formulated directly for lattice gauge fields, rather than for scalar fields. The natural observables are gauge-invariant quantities such as plaquette variables, Wilson loops, and character observables. An exact scale-replacement argument should therefore expose a secant response Gram on the space of such gauge-invariant observables, while the atlas should be compatible with gauge orbits and with the active gauge-invariant operator subspaces selected by the finite-scale RG flow. A theorem of this type could organize the finite-cutoff complexity layer of a possible RG approach to Yang–Mills. It would still need to be combined with genuinely model-specific ingredients: construction of the continuum Euclidean field, Osterwalder–Schrader positivity and reconstruction, and a uniform exponential-clustering estimate for gauge-invariant correlations, which is the probabilistic manifestation of a mass gap (Jaffe and Witten, 2000; Chatterjee, 2018). Accordingly, pointwise complexity should be viewed only as a possible finite-cutoff organizing framework for parts of the RG analysis; it should not be presented as a standalone route to the Yang–Mills mass gap.

Bottom line. The common mechanism with nonlinear DNNs is exact finite-scale telescoping plus a response Gram plus a data-independent atlas. In DNNs this exposes learned feature-Gram compression. In nonlinear RG it exposes active relevant/marginal operator directions and separates them from below-resolution irrelevant directions. The theorem is therefore a rigorous finite-dimensional complexity layer; scaling-limit Gaussianity, nontrivial continuum interaction, and gauge-theoretic mass gaps require additional model-specific estimates.

4 Graph and Local-Time Applications

4.1 Random walk on graphs and Gaussian free field

Let $G = (V, E, c)$ be a finite connected weighted graph with symmetric conductances $c_{xy} = c_{yx} \geq 0$, where $c_{xy} > 0$ exactly when $\{x, y\} \in E$. Write

$$c_x := \sum_{y \in V} c_{xy}.$$

The weighted graph Laplacian acts on functions $f : V \rightarrow \mathbb{R}$ by

$$(Lf)(x) = \sum_y c_{xy}(f(x) - f(y)).$$

We consider the continuous-time random walk, started at the root v_0 , that jumps from x to y at rate c_{xy}/c_x . Its normalized local time is

$$L_t(x) = \frac{1}{c_x} \int_0^t \mathbf{1}\{X_s = x\} ds, \quad \tau(t) := \inf\{s \geq 0 : L_s(v_0) > t\}.$$

This is the normalization for which the Green function and the Ray–Knight identity take their cleanest form.

The pinned Gaussian free field with root v_0 is the centered Gaussian vector $\eta = \{\eta_x\}_{x \in V}$ such that $\eta_{v_0} = 0$ almost surely and whose covariance is the killed Green function

$$\mathbb{E}[\eta_x \eta_y] = \Gamma_{v_0}(x, y) := \mathbb{E}_x[L_{T_{v_0}}(y)],$$

where T_{v_0} is the hitting time of v_0 . Equivalently, this covariance is the inverse of the pinned Laplacian on $V \setminus \{v_0\}$. Its canonical metric is

$$d(x, y) = (\mathbb{E}|\eta_x - \eta_y|^2)^{1/2} = \sqrt{R_{\text{eff}}(x, y)},$$

where R_{eff} is the effective-resistance metric associated with the conductance network.

4.2 Ray–Knight bridge and high-probability analysis

We now state the graph-theoretic consequences as applications of the Gaussian theorem. The walk is started at the root v_0 . Let $G = (V, E, c)$, L_t , $\tau(t)$, η , and η' be as in the preceding subsection.

We use the following finite-graph form of the second Ray–Knight theorem (Dynkin, 1983).

Theorem 4.1 (Second Ray–Knight isomorphism on a finite graph). *With the above normalization, under the product law of the walk and the independent fields (η, η') ,*

$$\left(L_{\tau(t)}(x) + \frac{1}{2}\eta_x^2\right)_{x \in V} \stackrel{d}{=} \left(\frac{1}{2}(\eta'_x + \sqrt{2t})^2\right)_{x \in V}. \quad (4.1)$$

In particular, $L_{\tau(t)}(v_0) = t$ almost surely.

The pointwise Gaussian theorem, combined with Theorem 4.1, yields a field-level envelope for local times. The independent field η' is essential in the proof because (4.1) is an equality in distribution for the enlarged field $L_{\tau(t)} + \eta^2/2$, not a pathwise identity for $L_{\tau(t)}$ alone. After the two Gaussian copies are controlled by the same pointwise envelope, the final statement can be expressed purely under the law of the walk.

Corollary 4.2 (Pointwise local-time envelope). *Let*

$$d(x, y) = (\mathbb{E}|\eta_x - \eta_y|^2)^{1/2} = \sqrt{R_{\text{eff}}(x, y)},$$

and let $A_\mu(x; \delta)$ be the envelope from (1.5) applied to the pinned GFF. Then, under the law of the walk, with probability at least $1 - \delta$,

$$\forall x \in V : \quad [t - \sqrt{2t} A_\mu(x; \delta/2)]_+ \leq L_{\tau(t)}(x) \leq t + \sqrt{2t} A_\mu(x; \delta/2) + \frac{1}{2} A_\mu(x; \delta/2)^2. \quad (4.2)$$

In particular, under the small-envelope condition

$$\sup_{x \in V} A_\mu(x; \delta/2) \leq \varepsilon \sqrt{t} \quad \text{for some } \varepsilon < 1/\sqrt{2},$$

which is the regime used in the cover- and blanket-time threshold criterion below, the lower bound in (4.2) is genuinely nontrivial:

$$[t - \sqrt{2t} A_\mu(x; \delta/2)]_+ = t - \sqrt{2t} A_\mu(x; \delta/2) > 0 \quad \forall x \in V.$$

The local-time envelope immediately yields the following blanket- and cover-time threshold criterion.

Corollary 4.3 (Blanket- and cover-time threshold criterion). *Assume that for some $t > 0$ and $\varepsilon \in (0, 1/\sqrt{2})$,*

$$\sup_{x \in V} A_\mu(x; \delta/2) \leq \varepsilon \sqrt{t}.$$

Then, under the law of the walk, with probability at least $1 - \delta$,

$$(1 - \sqrt{2}\varepsilon)t \leq L_{\tau(t)}(x) \leq \left(1 + \sqrt{2}\varepsilon + \frac{1}{2}\varepsilon^2\right)t \quad \forall x \in V.$$

In particular,

$$\tau_{\text{cov}} \leq \tau(t)$$

on this event, and

$$\max_{u,v \in V} \frac{L_{\tau(t)}(u)}{L_{\tau(t)}(v)} \leq \Delta_\varepsilon := \frac{1 + \sqrt{2}\varepsilon + \varepsilon^2/2}{1 - \sqrt{2}\varepsilon}.$$

Thus the normalized local-time field is within the multiplicative blanket factor Δ_ε of being flat.

The second Ray–Knight theorem is the precise field-level statement linking random-walk local times and quadratic Gaussian observables. It is important to retain the independent-copy formulation in Theorem 4.1. The left-hand side of (4.1) contains an independent quadratic field, and only the full shifted-square field has the law stated in the theorem. Thus, introducing the independent-copy field η' and formulating the probability event under the joint law of (η, η', X) are essential for a rigorous high-probability analysis of the local-time field.

Proof of Corollary 4.2. Set

$$A_x := A_\mu(x; \delta/2), \quad U_x := L_{\tau(t)}(x) + \frac{1}{2}\eta_x^2, \quad V_x := \frac{1}{2}(\eta'_x + \sqrt{2t})^2.$$

By Theorem 4.1,

$$(U_x)_{x \in V} \stackrel{d}{=} (V_x)_{x \in V}.$$

Define

$$D_x := \frac{1}{2}(\sqrt{2t} - A_x)_+^2, \quad B_x := t + \sqrt{2t}A_x + \frac{1}{2}A_x^2.$$

By Theorem 1.1 applied to the independent copy η' , the event

$$E' := \{\forall x \in V : |\eta'_x| \leq A_x\}$$

has probability at least $1 - \delta/2$. On E' , we have for every $x \in V$,

$$D_x \leq V_x \leq B_x.$$

Hence, by equality in distribution,

$$\mathbb{P}(\forall x \in V : D_x \leq U_x \leq B_x) \geq 1 - \delta/2.$$

Call this event F .

Next, applying Theorem 1.1 to η itself, the event

$$E := \{\forall x \in V : |\eta_x| \leq A_x\}$$

has probability at least $1 - \delta/2$. Therefore

$$\mathbb{P}(E \cap F) \geq 1 - \delta.$$

On $E \cap F$, for every $x \in V$,

$$L_{\tau(t)}(x) = U_x - \frac{1}{2}\eta_x^2 \leq B_x,$$

and also

$$L_{\tau(t)}(x) \geq D_x - \frac{1}{2}A_x^2.$$

Since local times are nonnegative, this gives

$$L_{\tau(t)}(x) \geq \max\left\{0, D_x - \frac{1}{2}A_x^2\right\}.$$

A direct case split on whether $A_x \leq \sqrt{2t}$ or $A_x > \sqrt{2t}$ shows that

$$\max\left\{0, D_x - \frac{1}{2}A_x^2\right\} = [t - \sqrt{2t} A_x]_+.$$

Thus, on an event of probability at least $1 - \delta$,

$$\forall x \in V : \quad [t - \sqrt{2t} A_x]_+ \leq L_{\tau(t)}(x) \leq t + \sqrt{2t} A_x + \frac{1}{2}A_x^2.$$

Since the event in (4.2) depends only on the walk, the same probability bound holds under the walk law. $\square$

Proof of Corollary 4.3. Under the assumption

$$\sup_{x \in V} A_\mu(x; \delta/2) \leq \varepsilon \sqrt{t} \quad \text{with} \quad \varepsilon < 1/\sqrt{2},$$

Corollary 4.2 gives, simultaneously for all $x \in V$,

$$L_{\tau(t)}(x) \geq [t - \sqrt{2t} \varepsilon \sqrt{t}]_+ = (1 - \sqrt{2} \varepsilon)t,$$

and

$$L_{\tau(t)}(x) \leq t + \sqrt{2t} \varepsilon \sqrt{t} + \frac{1}{2} \varepsilon^2 t = \left(1 + \sqrt{2} \varepsilon + \frac{1}{2} \varepsilon^2\right)t.$$

This proves the two-sided estimate.

Because the lower bound is strictly positive, every vertex has positive local time by time $\tau(t)$, hence every vertex has been visited and $\tau_{\text{cov}} \leq \tau(t)$. Finally, for any $u, v \in V$,

$$\frac{L_{\tau(t)}(u)}{L_{\tau(t)}(v)} \leq \frac{(1 + \sqrt{2} \varepsilon + \varepsilon^2/2)t}{(1 - \sqrt{2} \varepsilon)t} = \Delta_\varepsilon,$$

which is exactly the stated blanket estimate. $\square$

4.3 Classical cover-time theory and relation to the present result

For a finite connected conductance graph $G = (V, E, c)$, let τ_{cov} be the first time every vertex has been visited, and write

$$t_{\text{cov}}(G) := \max_{v \in V} \mathbb{E}_v[\tau_{\text{cov}}]. \quad (4.3)$$

A blanket time strengthens cover time by requiring that the walk has accumulated roughly stationary proportions of visits at all vertices. One convenient strong version leads to

$$t_{\text{bl}}(G, \delta) := \max_{v \in V} \mathbb{E}_v[\tau_{\text{bl}}(\delta)]. \quad (4.4)$$

Let $H(u, v)$ be the expected hitting time from u to v , and define the commute-time metric

$$\kappa(u, v) := H(u, v) + H(v, u).$$

Let $\mathcal{C} := \sum_{x \in V} c_x$ be the total conductance. The commute-time identity gives

$$\kappa(u, v) = \mathcal{C} R_{\text{eff}}(u, v).$$

For an unweighted graph, $\mathcal{C} = 2|E|$. If $\eta = \{\eta_v\}_{v \in V}$ denotes the pinned GFF, then $\mathbb{E}(\eta_u - \eta_v)^2 = R_{\text{eff}}(u, v)$.

Theorem 4.4 (Ding–Lee–Peres (Ding et al., 2012)). *For any connected conductance graph $G = (V, E, c)$ and any fixed $0 < \delta < 1$,*

$$t_{\text{cov}}(G) \asymp \gamma_2(V, \sqrt{\kappa})^2 = \mathcal{C} \cdot \gamma_2(V, \sqrt{R_{\text{eff}}})^2 \asymp_{\delta} t_{\text{bl}}(G, \delta). \quad (4.5)$$

Equivalently,

$$t_{\text{cov}}(G) \asymp \mathcal{C} \cdot \left(\mathbb{E} \max_{v \in V} \eta_v \right)^2. \quad (4.6)$$

Relation to the present result. Theorem 4.4 identifies the expected cover time with the square of the global GFF supremum scale, equivalently with Talagrand’s γ_2 functional. The present paper proves a different, prior field-level statement. Corollary 4.2 controls the whole local-time field before passing to a global threshold. Corollary 4.3 then recovers cover-time and blanket-time consequences only after one imposes a small-envelope condition. Thus the result here is complementary to the Ding–Lee–Peres theorem: it is a local-time and field-envelope statement, not a replacement for their global cover-time theorem.

5 Scope and Outlook

The paper has two finite-scale messages.

First, for Gaussian processes, the natural field-level refinement of classical generic chaining is not merely a scalar estimate on $\mathbb{E} \sup_{x \in T} X_x$, but a simultaneous pointwise envelope for the whole field. Theorem 1.1 provides such an envelope, with a local Fernique–Talagrand complexity and a variance-sensitive Gaussian tail term. After optimizing over the prior, its global in-expectation consequence recovers the anchored Fernique–Talagrand scale up to universal constants. For pinned Gaussian free fields, the same envelope transfers through the Ray–Knight isomorphism into high-probability local-time control before one thresholds the field to obtain cover-time and blanket-time consequences.

Second, the analogy between deep feature learning and renormalization should be understood as a proof architecture, not as a slogan. In both settings, the useful finite-scale analysis avoids infinitesimal Taylor expansion. One uses exact telescoping to replace one layer, or one RG scale, at a time. This exposes a response Gram matrix: the learned feature Gram in the DNN setting and the secant response Gram in the RG setting. The resulting pointwise ellipsoidal metric is then controlled by a hierarchical prior, whose cost decomposes into a local effective dimension and a global atlas term. In this sense, the finite-scale RG theorem identifies the correct analogue of the DNN local-chart/global-atlas mechanism.

The nonlinear RG results in this paper are deliberately finite-dimensional and finite-cutoff. They provide a structural pointwise-complexity theorem once a model-specific secant-Gram domination and phase/subspace atlas are available. They do not, by themselves, a complete route to

prove continuum interacting QFT, four-dimensional marginal triviality, or the Yang–Mills mass gap. Those problems require additional estimates: contraction of irrelevant or non-Gaussian remainders, Gaussianization or Wick-factorization mechanisms, gauge-invariant correlation decay, and, in the Yang–Mills case, a continuum construction with Osterwalder–Schrader positivity. The contribution here is to isolate a reusable finite-scale complexity layer that can be combined with such model-specific RG or correlation estimates.

Thus the Gaussian theorem can serve as a standard reference whenever one needs a field-level refinement of classical generic chaining, while the RG theorem clarifies what a pointwise-complexity approach to nonlinear finite-cutoff renormalization should control: the active response-Gram directions and the phase/subspace atlas cost, rather than a global supremum or a purely scalar Lipschitz bound.

References

- David Aldous. An introduction to covering problems for random walks on graphs. *Journal of Theoretical Probability*, 1989.
- Roland Bauerschmidt, David C. Brydges, and Gordon Slade. Introduction to a Renormalisation Group Method. *Lecture Notes in Mathematics*, vol. 2242, Springer, 2019.
- David C. Brydges and Gordon Slade. A renormalisation group method. V. A single renormalisation group step. *Journal of Statistical Physics*, 159:589–667, 2015.
- Sourav Chatterjee. Yang–Mills for Probabilists. arXiv:1803.01950, 2018.
- George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2:303–314, 1989.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989.
- Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. Clay Mathematics Institute Millennium Prize Problem description, 2000.
- Michael Aizenman and Hugo Duminil-Copin. Marginal triviality of the scaling limits of critical 4D Ising and ϕ_4^4 models. *Annals of Mathematics*, 194(1):163–235, 2021.
- I. M. Suslov. Is ϕ^4 theory trivial? arXiv:0806.0789, 2008.
- Ulli Wolff. Triviality of four dimensional ϕ^4 theory on the lattice. *Scholarpedia*, 2014.
- Marie Chupeau, Olivier Bénichou, and Raphaël Voituriez. Cover times of random searches. *Nature Physics*, 2015.
- Stéphane Condamin, Olivier Bénichou, Vicente Tejedor, Raphaël Voituriez, and Joseph Klafter. First-passage times in complex scale-invariant media. *Nature*, 2007.
- Amir Dembo, Yuval Peres, Jay Rosen, and Ofer Zeitouni. Cover times for Brownian motion and random walks in two dimensions. *Annals of Mathematics*, 2004.
- Jian Ding, James R. Lee, and Yuval Peres. Cover times, blanket times, and majorizing measures. *Annals of Mathematics*, 2012.

- E. B. Dynkin. Local times and quantum fields. *Seminar on Stochastic Processes*, 1983.
- X Fernique. Regularite des trajectoires des fonctions aleatoires gaussiennes. *Ecole d'Eté de Probabilités de Saint-Flour IV*, 1974.
- Aditya Grover and Jure Leskovec. node2vec: Scalable feature learning for networks. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.
- Shaojie Li and Yunbei Xu. Pointwise generalization in deep neural networks. URL: https://yunbeixu.github.io/assets/css/Pointwise_Generalization_in_Deep_Neural_Networks.pdf, 2026.
- nLab contributors. Constructive quantum field theory. nLab, 2026.
- Michel Talagrand. Regularity of Gaussian processes. *Acta Mathematica*, 1987.
- Herbert S. Wilf. The white screen problem. *Journal of Graph Theory*, 13(3):317–320, 1989.
- Yunbei Xu and Assaf Zeevi. Towards problem-dependent optimal learning rates. *Advances in Neural Information Processing Systems*, 2020.
- Yunbei Xu and Assaf Zeevi. Towards optimal problem dependent generalization error bounds in statistical learning theory *Mathematics of Operations Research*, 2025.